

MARS CODE MANUAL

VOLUME II: Input Requirements

Thermal-Hydraulic Safety Research Department

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PREFACE

Within the nuclear industry, there has been a strong demand for realistic analysis tools based on best-estimate modeling for application in the thermal hydraulic analyses of nuclear reactor systems. This demand has led to the development of such advanced codes as RELAP5, COBRA-TF, TRAC, CATHARE, etc.

Korea Advanced Energy Research Institute (KAERI) conceived and started the development of MARS code with the main objective of producing a state-of-the-art realistic thermal hydraulic systems analysis code with multi-dimensional analysis capability. MARS achieves this objective by very tightly integrating the one dimensional RELAP5/MOD3 with the multi-dimensional COBRA-TF codes.

RELAP5 is a versatile and robust code based on one-dimensional two-fluid formulation, and the code includes many generic component and special process models. COBRA-TF is based on three-dimensional two-fluid, three-field formulation and was developed especially for the reflood thermal-hydraulics of large-break Loss of Coolant Accident (LOCA).

The method of integration of the two codes is based on the dynamic link library techniques, and the system pressure equation matrices of both codes are implicitly integrated and solved simultaneously. In addition, the Equation-Of-State (EOS) for the light water was unified by replacing the EOS of COBRA-TF by that of the RELAP5.

Other developmental objectives are to modernize the coding and to provide some user convenience by providing simple but effective Graphic User Interface (GUI). With some key programming techniques dating back to 1960's, the coding of RELAP5 lags behind the current level of the computer industry. By taking advantages of structured programming techniques provided in modern compiler, FORTRAN 90, the RELAP part of the MARS code was thoroughly re-structured from the original RELAP coding with the objectives of improving the legibility, maintenance and ease of modification. The input scheme was unified whereby the original formatted input scheme of the COBRA-TF has been replaced with the free-format scheme of the RELAP. By this unification, a single free format input file is used for the two parts, one-dimensional and multi-dimensional. Some simple GUI are provided for the program execution and for the on-line showing of the calculation results. The MARS code has been successfully developed and it is being used in many key areas of the nuclear industry.

This input manual provides a complete list of input required to run MARS. The manual is divided largely into two parts, namely, the one-dimensional part and the multi-dimensional part. The inputs for auxiliary parts such as minor edit requests and graph formatting inputs are shared by the two parts and as such mixed input is possible. The overall structure of the input is modeled on the structure of the RELAP5 and as such the layout of the manual is very similar to that of the RELAP. This similitude to RELAP5 input is intentional as this input scheme will allow minimum modification between the inputs of RELAP5 and MARS. MARS development team would like to express its appreciation to the RELAP5 Development Team and the USNRC for making this manual possible.

**RELAP5/MOD3.3 CODE MANUAL
VOLUME II:
APPENDIX A
INPUT REQUIREMENTS**

Nuclear Safety Analysis Division

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1 INTRODUCTION

This volume completely describes data deck organization and data card requirements for all problem types allowed in MARS. The card numbers are in numerical order.

1.1 Control Format

Input is described in terms of input records or cards, where an input record or card is an 80-character record. Punched cards are nearly obsolete and one would be hard-pressed to find a key punch machine at most installations. Now, data are normally entered from interactive terminals, personal computers, or workstations, and the input usually exists only as disk files or is archived on tape. Data are usually viewed as lines on a CRT screen or lines of printed output. Nonetheless, the word card is used extensively in this input description to mean an input record.

RELAP5 attempts to read a 96-character record. If the actual input record is smaller, blank characters are added to the end of the input record to extend it to 96 characters. Each 96-character input record, preceded by a sequential card number starting at 1 and incrementing by 1, is printed as the first part of a problem output. Only the first 80 characters are used for MARS input; the additional 16 columns are for use with editors or utility programs such as UPDATE.

Most interactive editors allow the input of at least 80 character records. With many terminals allowing only 80 characters per line, it is convenient to limit the data record to 72 characters so that the data and editor supplied line numbers fit on one line (eight columns for line number and separator, 72 columns of data). Some editors provide for the optional storing of editor line numbers following the data portion of the record. If the data field is 72 columns, the line numbers might be stored in columns 73 to 80. These line numbers will be processed by RELAP5 as input, since RELAP5 uses the first 80 characters. To avoid this, either request the editor to store line numbers starting at character position 81, put a terminating character before the line number, or don't store the line numbers. The line numbers, if saved, are listed in the output echo of the input data.

If the UPDATE program is used to maintain the input deck, the update command must be used to specify that the card data are 80 columns instead of the default of 72.

1.2 Data Deck Organization

A RELAP5 problem input deck consists of at least one title card, optional comment cards, data cards, and a terminator card. A list of these input cards is printed at the beginning of each RELAP5 problem. The order of the title, data, and comment cards is not critical except that only the last title card and, in the case of data cards having duplicate data card numbers, only the last data card is used. We recommend that for a base deck, the title card be first, followed by data cards in card number order. Comment cards should be used freely to document the input. For parameter studies and for temporary changes, a new title card with the inserted, modified, and deleted data cards and identifying comment cards should be placed just ahead of the terminating card. In this manner, a base deck is maintained, yet changes are easily made.

When card format punctuation errors, such as an alphanumeric character in numeric fields are detected, a line containing a caret (^) located under the character causing the error and a message giving the card column of the error are printed. An error flag is set such that input processing continues, but the RELAP5 problem is terminated at the end of input processing. A standard RELAP5 error message (error message preceded by eight asterisks (*****)) is printed if a card error is found. Usually a card error will cause additional error comments to be printed during further input processing when the program attempts to process the erroneous data.

1.3 Title Card

A title card must be entered for each RELAP5 problem. A title card is identified by an equal sign (=) as the first nonblank character. The title (remainder of the title card) is printed as the second line of the first page following the list of input data. If more than one title card is entered, the last one entered is used.

1.4 Comment Cards

An asterisk (*) or a dollar sign (\$) appearing as the first nonblank character identifies the card as a comment card. Blank cards are treated as comment cards. The only processing of comment cards is the printing of their contents. Comment cards may be placed anywhere in the input deck except before continuation cards.

1.5 Data Cards

Data cards may contain varying numbers of fields that may be integer, real (floating point), or alphanumeric. Blanks preceding and following fields are ignored.

The first field on a data card is a card identification number that must be an unsigned integer. The value for this number depends upon the data being entered and will be defined for each type. If the first field has an error or is not an integer, an error flag is set. Consequently, data on the card are not used, and the card will be identified by the card sequence number in the list of unused data cards. After each card number and the accompanying data are read, the card number is compared to previously entered card numbers. If a matching card number is found, the data entered on the previous card are replaced by data from the current card. If the card being processed contains only a card number, the card number and data from the last previous card with that card number are deleted. Deleting a nonexistent card is not considered an error. If a card causes replacement or deletion of data, a statement is printed indicating that the card is a replacement card.

NOTE: For new problems, the recommended approach for deleting a model feature in the calculation input is to remove the appropriate cards or "comment" them out. Use of the "card number only" method mentioned above is not recommended. For restart problems, the recommended approach for deleting a model feature is to use the "delete" word that is typically input on the first data card for a model feature. The "card number only" method works only if the input cards previously appear in the restart input deck,

and in that is the case, it is recommended that the user either remove the appropriate cards or "comment" them out. In conclusion, the "card number only" method is not recommended for any problem.

Comment information may follow the data fields on any data card by beginning the comment with an asterisk (*) or dollar (\$) sign.

A numeric field must begin with either a digit (0 through 9), a sign (+ or -), or a decimal point (.). A comma or blank (with one exception, subsequently noted) terminates the numeric field. The numeric field has a number part and optionally an exponent part. A numeric field without a decimal point or an exponent is an integer field; a number with either a decimal point, an exponent, or both is a real field. A real number without a decimal point (i.e., with an exponent) is assumed to have a decimal point immediately in front of the first digit. The exponent part denotes the power of ten to be applied to the number part of the field. The exponent part has an E or e or D or d, a sign (+ or -), or both followed by a number giving the power of ten. These rules for real numbers are identical to those for entering data in FORTRAN E or F fields except that no blanks (with one exception) are allowed between characters to allow real data written by FORTRAN programs to be read. The exception is that a blank following an E or e or D or d denoting an exponent is treated as a plus sign. Acceptable ways of entering real numbers, all corresponding to the quantity 12.45, are illustrated by the following eight fields:

12.45, +12.45, 0.1245+2, 1.245+1, 1.245E 1, 1.245D+1, 1.245e 1, 1.245e+1

When entering a decimal zero for either an integer or floating point quantity, the 0 can be written in either form. Thus a floating point zero can be entered simply as 0 without a decimal point.

Alphanumeric fields have three forms. The most common alphanumeric form is a field that begins with a letter and terminates with a blank, a comma, or the end of the card. After the first alphabetic character, any characters except commas and blanks are allowed. The second form is a series of characters delimited by quotes (") or apostrophes ('). Either a quote or an apostrophe initiates the field, and the same character terminates the field. The delimiters are not part of the alphanumeric word. If the delimiter character is also a desired character within the field, two adjacent delimiting characters are treated as a character in the field. The third alphanumeric form is entered as nHz, where n is the number of characters in the field, and the field starts at the first column to the right of H and extends for n columns. With the exception of the delimiters (even these can be entered if entered in pairs), the last two alphanumeric forms can include any desired characters. In all cases, the maximum number of alphanumeric characters that can be stored in a word is eight. If the number of characters is less than eight, the word is left justified and padded to the right with blanks. If more than eight characters are entered, the field generates as many words as needed to store the field, eight characters per word, and the last word is padded with blanks as needed. Regardless of the alphanumeric type, at least one blank or comma must separate the field from the next field.

Most computers (e.g., workstations, CRAY, and IBM) hold only eight characters per word. All alphanumeric words required by RELAP5, such as components types, system names, or processing options, have thus been limited to eight characters. We highly recommend that the user limit all other one-

word alphanumeric quantities to eight characters so that input decks can be easily used on all computer versions. Examples of such input are alphanumeric names entered to aid identification of components in output edits.

1.6 Continuation Cards

A continuation card, indicated by a plus sign (+) as the first nonblank character on a card, may follow a data card or another continuation card. Fields on each card must be complete, that is, a field may not start on one card and be continued on the next card. The data card and each continuation card may have a comment field starting with an asterisk (*) or dollar (\$) sign. No card number field is entered on the continuation card, since continuation cards merely extend the amount of information that can be entered under one card number. Deleting a card deletes the data card and any associated continuation cards.

1.7 Terminator Cards

The input data are terminated by a slash or a period card. The slash and period cards have a slash (/) and a period (.), respectively, as the first nonblank character. Comments may follow the slash and period on these cards.

When a slash card is used as the problem terminator, the list of card numbers and associated data used in a problem is passed to the next problem. Cards entered for the next problem are added to the passed list or act as replacement cards, depending on the card number. The resulting input is the same as if all previous slash cards were removed from the input data up to the last period card or the beginning of the input data.

When a period card is used as the problem terminator, all previous input is erased before the input to the next problem is processed.

1.8 Sequential Expansion Format

Several different types of input are specified in sequential expansion format. This format consists of sets of data, each set containing one or more data items followed by an integer. The data items are the parameters to be expanded, and the integer is the termination point for the expansion. The expansion begins at 1 more than the termination point of the previous set and continues to the termination point of the current set. For the first set, the expansion begins at 1. The termination points are generally volume, junction, or mesh point numbers, and always form a strictly increasing sequence. The input description will indicate the number of words per set (always at least two) and the last terminating point. The terminating point of the last expansion set must equal the last terminating point. Two examples are given.

The first example is for the volume flow areas in a pipe component; the format is two words per set in sequential expansion format for *nv* sets. Using the number of volumes in the pipe (*nv*) as 10, the volume flow areas could be entered as

0010101 0.01,10 .

In this case, the volume flow areas for volumes 1 through 10 have the value 0.01.

The second example shows how the pipe volume friction data could be input. The input consists of three words per set for nv sets. The three words designate the wall roughness, hydraulic diameter (input of 0 causes the code to calculate it), and volume number. Possible data might be

0010801 1.0-6,0,8 1.0-3,0,9

0010802 1.0-6,0,10 .

Here, volumes 1 through 8 and 10 have the same values, and volume 9 has a different value.

1.9 Upper/Lower Case Sensitivity

Historically, computer systems allowed only upper case alphabetic characters. Accordingly, the following input descriptions use upper case for required input, e.g., SNGLVOL, 1.25E5. Now, many systems have upper and lower case alphabetic characters, and some applications are case sensitive, others not. At the NRC, required input must be in lower case, and the user should check the requirements at other installations. At installations with both upper and lower case capability, there are utilities (tr) and editors that can easily switch alphabetic characters to the desired case.

1.10 Data Card Requirements

In the following description of the data cards, the card number is given with a descriptive title of the data contained on the card. Next, an explanation is given of any variable data that are included in the card number. Then, the order of the data, the type, and the description of the data item are given. The type is indicated by A for alphanumeric, I for integer, and R for real.

2 MISCELLANEOUS CONTROL

2.1 Developmental Model Control, 1

This card has been added to the code for the convenience of developers in testing model improvements or new models. This card is not a standard input feature of the code. The description of this card has been added to the input requirements because several laboratories are receiving test versions to assist in the development and testing of the code. Anyone using this card must realize that they are selecting experimental options still under development. Furthermore, these options may change more frequently than the revision of this input manual. Thus, before using the options, users should obtain the brief listing of current options from the code (described below) and verify those descriptions against this manual.

The purpose of this card is to allow developers and analysts to quickly test new models by activating or deactivating a model through simple input instead of program modification, compilation, and loading. Ninety logical variables having only false or true values are provided and defined at the start of program execution as false. This input sets the logical variables to true or resets them to false at the beginning of a new problem or at any restart. Fortran IF statements added as part of the experimental coding activate or deactivate models based on the values of the logical variables.

As described above, up to 90 options can be defined and the options are identified with a number from 1 through 90. Which options are defined and what they control are very much version dependent. The usual practice is to enter the option capability using a currently unused option number as the new model or improvement is first coded. During further development and testing, the model may change and the effect of the option can change in a manner ranging from large to subtle. When the model has been completed or even abandoned, the production version of the model is coded and the option capability is removed. The option number is then available for reuse with a completely different model. Thus, the options are version dependent as to what option numbers are in use, what models they control, and the particular features of the models. Accordingly, these options should be used only by those in direct contact with the developers.

Each current option is described below. In light of the discussion above, the user should verify that the code version being used corresponds to this description. Programmers using this option feature are asked to include coding that issues error messages when unused options are selected and to issue a brief statement of the purpose of selected options. Remember, however, that all coding associated with these options is experimental and these output conventions may not be thoroughly checked.

Up to 91 numbers consisting of 0 or any of the currently available option numbers may be entered on this card. A positive nonzero number, n , activates option n by setting the logical variable n to true; a negative nonzero number, $-n$, deactivates option n by setting the logical variable n to false. Attempting to activate an unused option is an error, and attempting to deactivate an already inactive option or an unused option is also an error. Adding options or deactivating options is allowed at a restart; the previously defined options will remain. The status of the options is printed in any NEW or RESTART problem containing this card or a RESTART problem in which the restart point had an option selected. The printout

includes a listing of the 90 option numbers and a false (option not selected) or a true (option selected) value plus the brief description of each selected option.

The number 0 is not an option number but may be entered to force the brief descriptions of all available options to be printed regardless of whether they are active. The 0 input should be used only once to observe the available options and then removed so that the list better emphasizes the selected options.

W1-91. Option number as described above.

- | | |
|-----------|--|
| Option 0. | This option lists out all the available options. |
| Option 1. | This option adds an explicit viscous-like term to the momentum equations in the nearlyimplicit scheme. This term is documented in the discussion of the semi-implicit scheme. |
| Option 2. | This option sets the interfacial mass transfer to 0 and the direct heating heat transfer coefficient to 105. This option simulates the no mass transfer case in which the two phases are in thermal equilibrium. |
| Option 4. | This option scales back the Courant limit to 75% of its calculated value so that the time step will always be less than or equal to 75% of the Courant limit. |
| Option 6. | This option outputs information to the screen and to the screen file every time dtstep is called. It is useful for determining why the code is cutting the time step. The volume number for the volume that has the limiting Courant time step and the maximum mass error for each time step is printed on one line. In addition, the pressure and void fraction for each of these volumes are printed. If the time step is cut, the reason for the cut is given in the right-hand column. |
| Option 8. | This option provides time step control based on the change in void fraction and is designed to limit the time step when the void fraction in any cell is decreasing rapidly such as during periods of condensation. The time step will be repeated if the decrease in void fraction in any hydrodynamic volume is considered too large. The time step will be repeated . |
| Option 9. | This option provides a transition to plug flow when the vapor velocity exceeds the criterion for transition from stratified flow in horizontal geometries. The need for this change occurs during the reflood phase in the horizontal core of the Hanford N-Reactor. The process tubes in the core region are long and have a small diameter. As subcooled coolant enters the process tubes, high condensation occurs that results in local depressurization and high steam velocities towards the condensation site. Instabilities occur that eventually cause the code to fail. This changes introduces a plug flow model that limits condensation to a value that is just large enough to condense all the vapor that |

can flow at the critical velocity, utilizing the full channel cross-sectional area. The critical velocity is the velocity large enough to cause transition from a stratified flow condition. The concept behind this model is that condensation lowers the local pressure and draws in steam which pushes the water into a plug. The area of the plug limits condensation and reduces steam flow which causes a return to stratified flow.

- Option 10. This option provides time step control based on change in pressure within a hydrodynamic volume. This change causes the code to repeat a time step if the change in pressure during a time step exceeds the old time value, the new time value, or 50,000 Pa. This time step control allows the pressure to change by no more than a factor of two during a time step. This change generally causes the code to run slower but more reliably. With this change activated, the code will more accurately track pressure waves and oscillations but may cause certain problems to run unacceptably slowly. This option has no effect if Option 8 is not selected.
- Option 12. This option provides a user controlled (on/off) water packer developed for horizontal reactors such as the Hanford N-Reactor. The interfacial friction coefficient term for the momentum equation, C_i , is adjusted as a function of vapor void fraction. For void fractions of 0.001 or less, C_i is forced to an arbitrarily large value of 1010. For void fractions greater than 0.01, the regularly calculated C_i is used. For void fractions between 0.001 and 0.01, a cubic interpolation scheme is used to adjust C_i between the calculated value and 1010. As a cell is calculated to fill with liquid and the calculated cell pressure rises, the lower inertia vapor is the phase first to respond, either moving on to the next cell or moving back to an upstream cell. The effect of this model is that, as the void fraction decreases, the interfacial drag is increased, thus allowing the moving vapor to either drag liquid on to the next cell or impede the incoming liquid from an upstream cell. In either case, the model eases the overfilling and overpressurization of the cell.
- Option 13. This option activates vertical stratification changes.
- Option 14. This option turns off constitutive relations and should only be used for testing advancement of the basic advancement scheme for two-phase conditions. Do not use for single phase conditions.
- Option 16. This option activates the velocity flip/flop logic for the nearly-implicit advancement scheme.
- Option 17. This option allows the code to run at the Courant limit with a multiplication factor of four instead of two and a reduction factor of 0.25 instead of 0.5 used in the mass error check for the time step control.
- Option 18. This option adds the sharp interface and reverse void profile logic from MOD2.5.

- Option 19. This option uses the Bestion correlation (in MOD2.5) rather than the EPRI correlation (in MOD3) for bundles (when volume flag $b = 1$).
- Option 20. This option uses the annulus multiplication factor for bubble rise velocity. This model change was raised due to downcomer boiling issue for LBLOCA of DVI plant.
- Option 21. Do not use. Presently option only acquires additional storage arrays for testing numerical techniques.
- Option 23. This option selects a boron transport algorithm (Godunov) that greatly reduces the numerical diffusion of boron compared to the standard algorithm.
- Option 24. This option activates the original RELAP5 subcooled boiling model. The default subcooled boiling model is the SRL model, which is better at low pressures.
- Option 25. This option linearizes the interfacial heat transfer for the nearly implicit scheme.
- Option 32. This option activates the stretch logic for water packing model.
- Option 36. Option 36 limits metastable extrapolation to 50 K.
- Option 37. This option turns off the umbrella model. When the umbrella model is on, an upper limit is placed on the liquid interfacial heat transfer coefficient (H_{if}) when the liquid is subcooled. The limit is umbrella shaped so as to force the coefficient to small values as the void fraction approaches 0.0 or 1.0.
- Option 41. This option includes energy dissipation due to form loss.
- Option 42. This option applies a stronger unchoking test for junctions with an abrupt area change.
- Option 43. This option uses iteration to calculate the hydrodynamic conditions at the throat for critical flow of vapor.
- Option 45. This option selects the newly developed model for condensing interphase heat transfer.
- Option 48. This option requests use of the level tracking model during moving problems.
- Option 50. This option activates at all junctions where choking option is activated the original RELAP5 critical flow model instead of the default Henry-Fauske critical flow model. The code does not allow a user to switch between the two critical flow models in the middle of a problem, e.g. switching from the Henry-Fauske to the original critical flow model after a steady-state run at the beginning of a transient run.

- Option 51. Normally, water packing is activated in all volumes unless specifically disabled by an input volume flag. This option disables water packing for all volumes.
- Option 52. Normally, the choking model is activated for all junctions unless specifically disabled by an input junction flag. This option disables the choking model for all junctions.
- Option 53. This option activates the Moody critical flow model. It is part of the CANDU package.
- Option 54. This option activates the M.S.Chung modified mechanical critical flow model.
- Option 55. This option activates the HF critical flow based on Enthalpy Upstream.
- Option 56. This option enforces $v_g = v_f$ at $g = 1$ in BubbleDropDrag(FIDIS2).
- Option 57. Turbulent viscous stress terms in 3D channel and gap module is activated.
- Option 61. This option activates the Podowski subcooled boiling model instead of the Lahey method.
- Option 62. This option uses newly developed changes to the Chen F factor in PREDNB to minimize oscillations.
- Option 64. This option uses the old halving and doubling of the time step but the new Courant limit.
- Option 65. This option changes the subcooled boiling model by modifying the fraction of nucleate boiling heat flux that generates vapor when the bulk liquid is subcooled. The modification minimizes the “on/off” behavior associated with low-pressure/low-flow conditions.
- Option 66. This option implements donor/acceptor differencing in vertical stratification volumes.
- Option 67. This option activate the lateral momentum flux only for MULTID component. Using this option, the diffusion term of MULTID component is neglected.
- Option 68. This option use serial X-Y volume-junction ID number for MULTID Component
- Option 69. This option causes Two Step Method using IPF Model. Virtual Mass Force is deactivated
- Option 70. This option use ASME 67 type Dynamic property instead of ASME 93 property.
- Option 71. This option turns on the TMDPVOL momentum flux term.
- Option 75. This option causes the vaporization smoothing.
- Option 76. This option causes the default junction diameter as a min. volume hydraulic diameter.

- Option 77. This option activates the SNU bubble regime model in annulus.
- Option 78. This option deactivates 3D water-packing scheme.
- Option 80. This option enables the moving reactor model by KAERI
- Option 81. This option activates the dryout model using critical thickness correlation.
- Option 82. This option activates 3D dial.
- Option 83. This option activates generation of MARS data file for interfacing with MIDAS severe accident code.
- Option 84. This option activates implicit gravity head and horizontal stratification force due to the void gradient in the nearly implicit scheme.
- Option 85. This option produce tecplt.dat file of MULTID output parameter for TECPLOT 3rd party graphic software.
- Option 86. This option deactivates option 60. It restores the old Courant time step calculation using void weighted volume averaged velocities with icoran = 1.
- Option 87. This option deactivates the NC gas equation in the 3D module and a minimum internal energy of gas phase in the 1D module is set to 2.3e6 J/kg.
- Option 88. uses K KRB1 CHF correlation at P>50 bar rather than AECL Lookup Table
- Option 89. uses KNOEBEL Annuli CHF correlation in annuli geometry (120)
- Option 90. This option turns on the e-flag (modified PV term in the energy equation) for all applicable junctions. This includes all junctions except for time-dependent junctions, accumulator junctions, and junctions associated with separator, jetmixer, pump, turbine, and eccmixer components.
- Option 99. This turns on Iterative Matrix solver for sparse matrix.

2.2 Multiplier for the Global Models, 10

This input card was added for implementing the multipliers for the two-phase wall friction and the interfacial drag models, Ishii-Mishima entrainment coefficient, and We number of reflood, which can be used for the uncertainty quantification of MARS code.

Up to 10 multipliers can be entered in Card 10 and, at present, only two multipliers are used. If Card 10 is not entered, the default multipliers 1.0 are used, which means no changes in the two-phase pressure drop and interfacial drag models.

W1(R)	Multiplier for the two-phase wall friction.
W2(R)	Multiplier for the interfacial drag model.
W3(R)	Multiplier for the Ishii-Mishima entrainment coefficient.
W4(R)	Multiplier for We number of reflood.

2.3 Multiplier for the Component-wise two phase wall friction, 11

This input card was added for implementing the multipliers for the two-phase wall friction for the specified components. Up to 9 components can be entered in card. If card is not entered, the default multipliers 1.0 are used.

W1(R)	Multiplier.
W2(I)	Component 1 identification number: CCC
W3(R)	Component 2 identification number: CCC
W4(R)	Component 3 identification number: CCC
...	
WN(I)	Component N identification number: CCC

2.4 Multiplier for the component-wise interfacial drag model, 12

This input card was added for implementing the multipliers for the interfacial drag model for the specified components. Up to 9 components can be entered in this card. If card is not entered, the default multipliers 1.0 are used.

W1(R)	multiplier.
W2(I)	Component 1 identification number: CCC
W3(R)	Component 2 identification number: CCC
W4(R)	Component 3 identification number: CCC

...

WN(I) Component N identification number: CCC

2.5 Multiplier for the component-wise interfacial heat transfer model, 13

This input card was added for implementing the multipliers for the interfacial heat transfer model for the specified components. Up to 9 components can be entered in this card. If card is not entered, the default multipliers 1.0 are used.

W1(R) multiplier.

W2(I) Component 1 identification number: CCC

W3(R) Component 2 identification number: CCC

W4(R) Component 3 identification number: CCC

...

WN(I) Component N identification number: CCC

2.6 Problem Type and Option, 100

This card is always required.

W1(A) Problem type. Enter one of the following: NEW, RESTART, RESET, REEDIT, STRIP, or CMPCOMS.

NEW specifies a new simulation problem.

RESTART specifies continuation from some point in a previous problem using information from the RSTPLT file.

STRIP specifies that data are to be extracted (stripped) from the RSTPLT file, and only the data specified are written to the STRIP file.

W2(A) Problem option. This word is needed if W1 is NEW or RESTART and is optional if W1 is STRIP.

If NEW or RESTART is entered, in W1, enter either STDY-ST or TRANSNT in W2 to specify the type of simulation. Note the cautions discussed in Section 2.9 when the problem option is changed from STDY-ST to TRANSNT or vice versa.

When STRIP is entered in W1, enter either BINARY or FMTOU or leave W2 blank. BINARY is assumed if W2 is not entered. BINARY indicates the unformatted (BUFFER OUT) file. FMTOU indicates that the same information is to be written as 80-column formatted records. One use of this option is to allow simulation results to be transmitted to a different type of computer. Formats are

STRIP Record 1. (5A8,10X,A8)

STRIP Record 2. (A10,3I10)

STRIP Record 3. (8A10)

STRIP Record 4. (A10,7I10/(8I10))

STRIP Record 5..., N. (A10, 5X,1P,4E15.6/(5E15.6)).

The STRIP record above refers to the data in one record of the unformatted file. Multiple 80-column formatted records may be written for STRIP Records 3 through N.

2.7 Input Check or Run Option, 101

This card is optional for all types.

W1(A) Option. Enter either INP-CHK or RUN; if this card is omitted, RUN is assumed. This card has no effect on a CMPCOMS problem

If INP-CHK is entered, the problem execution stops at the end of input processing.

If RUN is entered, the problem is executed if no input errors are detected.

2.8 Units Selection, 102

This card is optional for all problem types. If the card is omitted, SI units are assumed for both input and output. If the card is used, enter either SI or BRITISH for each word. SI units used are the basic units, kg, m, s, and the basic combined units such as $\text{Pa} = \text{kg} \cdot \text{m} / \text{s}^2 \cdot \text{m}^2$. British units are a mixture of lb (mass), ft, and s, primarily, but pressure is in $\text{lb}_f / \text{in}^2$ (lb_f is pounds force), heat energy is in Btu, and power is in MW. Thermal conductivity and heat transfer units use s, not h.

W1(A) Input units.

W2(A) Output units. If this word is missing, SI units are assumed for output.

2.9 Restart Input File Control, 103

This card is required for all problem types (W1 of card 100) except NEW and is not allowed for type NEW.

When the problem option (W2 on card 100) is the same as the problem being restarted, the steady-state or transient is continued, and data on the RSTPLT file up to the point of restart are saved. If the restart continues from the point the previous problem terminated, restart and plot information is added to the end of the previous RSTPLT file. If the restart is prior to the termination point of the previous simulation, restart and plot data after the point of restart are overwritten by new results. A copy should be saved if RSTPLT files from each simulation are needed. If the problem options are different, data up to the point of restart are not saved, problem advancement time is reset to 0, and the RSTPLT file will contain information as if this problem type were NEW.

Some cautions should be observed when the problem advancement time is changed by changing the problem option from STDY-ST to TRANSNT, or vice versa, or the problem advancement time is reset through W1 on card 200. Either or both of these could be specified at restart. When the advancement time is changed, the user is responsible for ensuring that models involving problem time will operate as intended. Affected models include trips using advancement time, control systems using time as an operand (does not include differentiation or integration with respect to time), and table lookup and interpolation using time as the independent variable. If necessary, trips, control systems, general tables, time-dependent volumes, junctions, and pump speed tables can all be reentered at restart. With normal modeling practices, little use of modeling features involving advancement time is needed for runs to steady-state and accordingly little effort should be needed in switching from STDY-ST to TRANSNT. Because of the frequent use of time in logic to initiate failures, as part of safety systems, and used in establishing the delay times allowed in most table lookup and interpolation tables, required changes to a transient run may be extensive.

The program does make a change to delay control components when the advancement time is changed. The delay control component operates by maintaining a tabular past history of the delayed functions and using table lookup and interpolating to evaluate the delayed function. The table consists of pairs of time values and the delayed function. When the problem time is changed, the time values in the history table and the time value to store the next point in the table are modified by adding the difference of the new advancement time and the old advancement time. The modified history table is as if the problem being restarted was run with the new advancement time. This may not be the desired change, and, in that case, the user can reenter the delay component.

W1(I) Restart number. This must be a number printed in one of the restart print messages and whose associated restart information is stored in the RSTPLT file. If the problem type (W1 on card 100) is STRIP, this number must be 0.

Alternately, a -1 may be specified instead of a specific restart number. This has the effect of restarting a problem from the last block in the restart file. Thus, the user can now set up a series of restart cases which begin from the end of a previous run without having to wait for the run to finish in order to get the actual restart number from which to restart.

- W2-6(A) Restart plot file name. This optional alphanumeric entry can be used to enter the file name of the restart plot file. Up to forty characters may be entered as one alphanumeric field. (The code internally treats the field as up to five eight-character words.) The default file name for the restart plot file is rstplt. This may be overridden on Unix machines by using the -r option on the command line. Either the default name, the name from the command field, or the name from this field on a previous case may be overridden by this field.

2.10 Restart-Plot File Control, 104

This card can be entered for NEW, RESTART, and STRIP options. For the strip option, this card controls the strip file, and the NONE option is not allowed. Only the FILENAME option can be used. If this card is omitted, the restart-plot file is rewound at the end of the problem, but no further action is taken. The user may need to provide system control cards to dispose of the file. To prevent the restart-plot file from being written, a card with NONE must be entered.

- W1(A) Action. This word may not be blank. If this word is NONE, no restart-plot file is written. If this word is FILENAME, words 2-6 on this card contain the restart plot file name.
- W2-6(A) Restart plot file name. This optional alphanumeric entry can be used to enter the file name of the restart plot file. Up to forty characters may be entered as one alphanumeric field. (The code internally treats the field as up to five eight-character words.) The default file name for the restart plot file is rstplt. This may be overridden on Unix machines by using the -r option on the command line. Either the default name, the name from the command field, or the name from this field on a previous case may be overridden by this field. This information can be entered only on NEW problems; in RESTART problems, this information may be entered on the 103 card.

2.11 CPU Time Remaining and Diagnostic Edit, 105

Card 105 controls termination of the transient advancement based on the CPU time remaining for the job. Some operating systems allow specification of the CPU time allocated for a job as part of the job control language and also provide a means to determine the CPU time remaining during job execution. As an alternative, Word 3 of this card may be entered as the CPU time allocated. An alternative CPU remaining time is computed by decrementing this quantity by the CPU used as measured by the program. If Word 3 is omitted or 0, the alternative CPU remaining time is assumed infinite. At the end of each time step, the CPU time remaining for the job is determined from the minimum of the system (if available) and alternative CPU remaining times. If the remaining CPU time is less than Word 1, the transient is immediately terminated. The advancement may not be at the end of a requested time step due to time step

reduction; the hydrodynamic, heat conduction, and reactor kinetics may not be advanced to the same point; or the advancement may not be successful and the advancement is scheduled to be repeated with reduced time step. Major edits, minor edits, plot edits, and a restart record are forced. The transient can be restarted from this point as if the problem had not been interrupted. The transient is also terminated after successful advancement over a requested time step and the CPU time is less than Word 2. Word 2 should be larger than Word 1. The default values for Words 1 and 2 are 1.0 and 2.0 seconds. The default values are used if the card is not supplied or the entered numbers are less than default values. Word 2 is also forced to be 1.0 seconds larger than Word 1. The time values must include time for the final minor and major edits (very little time required), plotting, and any other processing that is to follow termination of RELAP5 execution. This card is optional, but we strongly recommend its use with Word 3 nonzero on systems that do not provide a system CPU limit.

Card 105 also controls the diagnostic edit printout through the use of Words 4 and 5. If these words are missing or 0, no debug options are in effect. If Word 4 is greater than 0, then Word 4 is the attempted advancement count number to start a diagnostic edit, and Word 5 is the attempted advancement count number to stop the diagnostic edit as well as the calculation. If Word 4 is -1, a dump file is written on the file specified by the -A option on the command line at the completion of the advancement given in Word 5. Entering 0 in Word 5 writes the dump file just before the start of transient advancement. The problem is terminated after writing the dump file. If Word 4 is -2, a dump file is written on the file given by the -A option after the advancement given in Word 5; the time advancement is then repeated and a dump file following the repeated advancement is written on the file given by the -B option. The problem is terminated after writing the second dump file. Word 5 must be greater than 0 when Word 4 is -2. The default file names are -A dumpfil1 and -B dumpfil2.

W1(R)	CPU remaining limit 1 (s).
W2(R)	CPU remaining limit 2 (s).
W3(R)	CPU time allocated (s). This quantity is optional.
W4(I)	Debug control word as described above.
W5(I)	Debug control word as described above.

If the program is compiled with compile time option CTSS defined, entering Word 1 as 0.0 will cause no testing for CPU termination and normal CTSS termination at the end of CPU time can occur. In this case, the problem can be restarted.

2.12 Noncondensable Gas Species, 110

This card is required for all calculations that use noncondensable gas. Nitrogen must be included for any problem having accumulators. This card cannot be entered on a RESTART problem.

W1-WN(A) Noncondensable gas type. Enter any number N of words (maximum 5) of the following noncondensable gas types: ARGON, HELIUM, HYDROGEN, NITROGEN, XENON, KRYPTON, AIR or SF6.

2.13 Noncondensable Mass Fractions

Card 115 is related to card 110. card 115 is required if card 110 is entered unless only one species is entered on card 110, and then the mass fraction is set to 1.0. The number of words on card 115 must equal the number of words on card 110. The sum of the mass fractions must sum to 1.0. The mass fractions on these cards are default values and are used for initial conditions of active volumes and for values of time-dependent volumes. This card cannot be entered on a RESTART problem.

2.13.1 Initial Mass Fraction for Each Noncondensable Gas Type, 115

W1-WN(R) Mass fraction for each noncondensable gas type.

2.14 Card 119, Gravity Constant

This card is optional and specifies the gravitational constant. If not entered, the earth gravitational constant of 9.80665 m/s^2 is used.

W1(R) Gravitational constant (m/s^2 , ft/s^2). A positive number, which must be greater than or equal to $1.010 \times 10^{-6} \text{ m/s}^2$ (or $3.2808 \times 10^{-6} \text{ ft/s}^2$ if British input is used), is used as the gravitational constant. If -1.0 is entered, the earth gravitational constant is used.

2.15 Hydrodynamic System Control

Independent hydrodynamic systems can be described by the hydrodynamic component input. The term independent hydrodynamic systems means that there is no possibility of flow between the independent systems. A typical example would be the primary and secondary systems in a reactor where heat flows from the primary system to the secondary system in the steam generator but there is no fluid connection. If a tube rupture were modeled, the two systems would no longer be independent. Input processing lists an elevation for each volume in each independent hydrodynamic system and includes a check on elevation closure for each loop within a system. A reference volume is established for each system through input or default.

These cards are optional for each system. If not entered for a system, that system contains H_2O as the fluid unless a different fluid is specified in hydrodynamic component data, and the lowest numbered volume in each system is the reference volume. Additionally, the reference volume has a default elevation of 0. These cards should not be entered in a RESTART problem.

2.15.1 Hydrodynamic System, 120-129

- W1(I) Reference volume number of the system. This must be a volume in the hydrodynamic system.
- W2(R) Reference elevation of the volume center relative to a fixed z-axis for the system (m, ft).
- W3(A) Fluid type for the system. Enter H2O or H2ONEW or D2O or HE for helium or CO2 for carbon dioxide. If H2ONEW is entered, the code uses the newer steam tables and if H2O is entered, the code uses the older thermodynamically-consistent tables. The newer tables are based on the Helmholtz functional fit in the IAPWS-1995 document.
- W4(A) Optional alphanumeric name of system used in output editing. *NONE* is used if this word not entered.
- W5(I) System information flag. This word has the packed format g. This word is optional. If this word is not entered, g = 0 is used.
- The digit g specifies whether noncondensable gas is present for this system. g = 0 specifies that noncondensable gas is present for this system. g = 1 specifies that noncondensable gas is not present for this system. If g = 1 (no noncondensable) in a system and if the digit t = 4, 5, or 6 in the hydrodynamic volume component control word gbt, an input error will result.
- If the 99 option in CARD 1 is on, g = 2 specifies that noncondensable gas is present and the iterative scheme is used. g = 3 specifies that noncondensable gas is not present and the iterative scheme is used.
- W6-9(A) Thermodynamic property file name of the system. This optional alphanumeric entry can be used to enter the file name of the thermodynamic property file of the system. Up to thirty-two characters may be entered as one alphanumeric field. (The code internally treats the field as up to four eight-character words.) The default file name for the thermodynamic property file is tpfh2o for H2O, tpfd2o for D2O, tpfh2onew for H2ONEW, tpfhe for helium, tpfco2 for CO₂, etc.

2.16 Self-Initialization Option Control

These cards are optional, are not needed, and are only used as a cross-check on the controllers specified in Section 14. Data supplied on these cards are used to invoke the self-initialization option. These data describe which and how many of each controller will be used. To retain generality and flexibility, the self-initialization option does not require that the steady-state and nearly-implicit solution scheme options be concurrently turned on. However, this is the recommended procedure. These latter options are invoked

through input data cards 100 and 201 through 299. In addition to the data cards described below, the user must furnish data on the controllers to be used, as described in Section 14.

2.16.1 Self-Initialization Control, 140

This card specifies the number and type of controllers desired.

W1(I) Number of pump controllers.

W2(I) Number of steam flow controllers.

W3(I) Number of feedwater controllers.

or

W1(A) DELETE or DISCARD. The command DELETE is allowed only in RESTART problems, and self-initialization control must be existing at the time of restart. The DELETE command deletes all controllers.

2.16.2 Self-Initialization Pump Controller and Identification, 141-142

These cards establish the relationship between the pump number and the number of the pump controller. For each pump so referenced, the user *must* use the time-dependent pump velocity option. For pump component card CCC6100, Words 2 and 3 must be the alphanumeric and numeric parts for the pump controller. The time-dependent pump velocity data (pump component cards CCC6100 through CCC6199) should be input so that the search variable and pump velocity are related by a straight line through the origin with a slope of 1.

W1(I) Component number of pump number.

W2(I) Controller identification number for pump number 1.

W3(I) Component number of pump number 2.

W4(I) Controller identification number for pump number 2.

A maximum of six pump/controller pairs may be entered.

2.16.3 Self-Initialization Steam Flow Controller Identification, 143-144

These cards establish the relationship between the steam flow control valve number and the steam flow controller number.

W1(I) Component number of steam flow control valve number 1.

W2(I) Controller number of steam flow controller for steam flow control valve number 1.

W3(I) Component number of steam flow control valve number 2.

W4(I) Controller number of steam flow controller for steam flow control valve number 2.

A maximum of six control valve/controller pairs may be entered. Note that in the above the valve component is assumed to be the control component. However, the user is not constrained to use a valve and may use a pump or a time-dependent junction. CAUTION: only a servo valve, a time-dependent junction, or a pump may be used, or a diagnostic error will result.

2.16.4 Self-Initialization Feedwater Controller Identification, 145-146

These cards establish the relationship between the feedwater valve number and the feedwater controller number.

W1(I) Component number of feedwater valve number 1.

W2(I) Controller id number of the feedwater controller for feedwater valve number 1.

W3(I) Component number of feedwater valve number 2.

W4(I) Controller id number of the feedwater controller for feedwater valve number 2.

A maximum of six control valve/controller pairs may be entered. Note that in the above it is assumed that a valve component is the control component. However, the user is not constrained to use a valve and may use a pump or time-dependent junction. CAUTION: only a servo valve, time-dependent junction, or a pump is allowed, or a diagnostic will result, such as a time-dependent junction with the controller output used as the independent variable in place of time.

2.16.5 Pressure and Volume Control Component Identification, 147

This card identifies the component number, connection data, and pressure level for the time-dependent volume that is to provide pressure and volume control during the self-initialization null transient.

W1(I) Component number of time-dependent volume that replaces the pressurizer.

W2(I) Component number to which the above time-dependent volume is connected; CAUTION: only a single-junction is allowed or an error will result.

W3(R) Desired steady-state pressure.

3 TIME STEP CONTROL

3.1 Initial Time Value and User-Controlled Time Step, 200

This card is optional. See the description of each word on this card for the default values if this card is not entered.

- W1(R) Initial time. If not entered, the simulation time at the start of the advancements is 0 for a NEW problem, the advancement time at the point of restart for a RESTART problem, or 0 for a RESTART problem in which the problem option switches from STDY-ST to TRANSNT or vice versa. If this card is entered, the simulation time is set to the entered value, which must be greater than or equal to 0. Setting the simulation time with this entry can be done on any NEW or RESTART problem but with most applications should only be used in NEW or RESTART problems that switch from the STDY-ST or TRANSNT options. See the cautions discussed in Section 2.9 for this capability. When needing to enter W2 but do not wish to enter a new initial time, enter -1.0, which is a flag to ignore this word.
- W2(I) Control variable number for user-controlled time step. This word is optional. A nonzero number specifies a control variable whose value is used for user-specified time step control. The time step will be determined from the maximum of the value of the control variable and the current minimum time step entered on cards 201 through 299. The time step will be equal to or less than this value and depends on the current requested time step, the mass error and other error checks, the Courant limit, and the time-step reduction options.

3.2 Time Step Control, 201-299

At least one card of this series is required for NEW problems. If this series is entered for RESTART problems, it replaces the series from the problem being restarted. This series is not used for other problem types. Card numbers need not be consecutive.

- W1(R) Time end for this set (s). This quantity must increase with increasing card number. The default action for the code is to stop at the first time step that is greater than or equal to this time. This means the code will probably not stop at the end time specified on this card, but will stop at some time larger than the end time. If the user wants to stop exactly at the end time, then the user needs to prefix the end time with a negative (-) sign.
- W2(R) Minimum time step (s). The code starts out with this time step and increases by a factor 1.1 for the next step if the mass error is sufficiently low.
- W3(R) Maximum time step (s). This quantity is also called the requested time step.

W4(I) Control option. This word has the packed format ssdt. It is not necessary to input leading zeros.

ss. These digits represent a number from 0 through 15, are used to control the printed content of the major edits. The number is treated as a four-bit binary number. To decide how to set ss, pick some appropriate numbers from the set 0, 1, 2, 3, and 4, using the description below to decide which numbers to pick, then add the numbers together and set ss to that number.

ss = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), all the standard major printed output is given.

ss = 1. OMIT HEAT STRUCTURE TEMPERATURE BLOCK OUTPUT. If the first bit from the right is set (i.e., ss = 1 if the other bits are not set), the heat structure temperature block is omitted.

ss = 2. OMIT SECOND PORTION OF THE JUNCTION BLOCK OUTPUT. If the second bit from the right is set (i.e., ss = 2 if the other bits are not set), the second portion of the junction block is omitted.

ss = 4. OMIT THIRD AND FOURTH PORTIONS OF VOLUME BLOCK OUTPUT. If the third bit from the right is set (i.e., ss = 4 if the other bits are not set), the third and fourth portions of the volume block is omitted.

ss = 8. OMIT STATISTICS BLOCK. If the fourth bit from the right is set (i.e., ss = 8 if the other bits are not set), the statistics block is omitted.

d. This digit represents a number from 0 through 7, can be used to obtain extra output at every hydrodynamic time step. The number is treated as a three-bit binary number. To decide how to set d, pick some appropriate numbers from the set 0, 1, 2, and 3, using the description below to decide which numbers to pick, then add the numbers together and set d to that number.

d = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), the standard output at the requested frequency using the maximum time step is obtained (see words 5 and 6 of this card). If the number is nonzero, output is obtained at each successful time step; and the bits indicate which output is obtained. These options should be used carefully, since considerable output can be generated.

d = 1. MAJOR EDITS EVERY TIME STEP. If the first bit from the right is set (i.e., d = 1 if the other bits are not set), major edits are obtained every successful time step.

$\underline{d} = 2$. MINOR EDITS EVERY TIME STEP. If the second bit from the right is set (i.e., $\underline{d} = 2$ if the other bits are not set), minor edits are obtained every successful time step.

$\underline{d} = 4$. PLOT RECORDS EVERY TIME STEP. If the third bit from the right is set (i.e., $\underline{d} = 4$ if the other bits are not set), plot records are written every successful time step. As a practical example, assume you made a run with $\underline{d} = 0$ and discovered that the plot points were too far apart to tell what was happening in the transient. One solution would be to decrease the plot frequency W5, to 1 to get a plot point for each requested time step and decrease the maximum time step size to get the desired detail in the plot. An easier solution would be to set $\underline{d} = 4$ and get plot points out each successful time step, even if they are smaller than the requested time step. For example, if $\underline{tt} = 3$, then changing W4 from a 3 to a 403 would do the trick.

The digits $\underline{tt}$, that represent a number from 0 through 63, are used to control the time step. The number is treated as a six-bit binary number. The effect of no bits being set, i.e., 0 being entered, and the effect of each bit are first described followed by the recommended combination of bits. To decide how to set $\underline{tt}$, pick some appropriate numbers from the set 0, 1, 2, 4, 8, 16, and 32 using the description below to decide which numbers to pick, then add the numbers together and set $\underline{tt}$ to that number.

$\underline{tt} = 0$. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), no error estimate time step control is used, and the maximum time step is attempted for both hydrodynamic and heat structure advancement. The hydrodynamic time step, however, is reduced to the material Courant limit and further to the minimum time step for cases such as water property failures.

$\underline{tt} = 1$. MASS ERROR CONTROLS THE TIME STEP. If the first bit from the right is set (i.e., $\underline{tt} = 1$ if no other bits are set), the hydrodynamics advancement, in addition to the time step control when no bits are set, uses a mass error analysis to control the time step between the minimum and maximum time step.

$\underline{tt} = 2$. HYDRODYNAMIC TIME STEP SAME AS HEAT CONDUCTION TIME STEP. If the second bit from the right is set (i.e., $\underline{tt} = 2$ if the other bits are not set), the heat conduction/transfer time step is the same as the hydrodynamic time step; if the second bit from the right is not set, the heat conduction/transfer time step uses the maximum time step.

$\underline{tt} = 4$. HEAT CONDUCTION AND HYDRODYNAMICS COUPLED IMPLICITLY. If the third bit from the right is set (i.e., $\underline{tt} = 4$ if the other bits are not set), the heat conduction/transfer and hydrodynamics are coupled implicitly; if the third bit from the right is not set, the heat conduction/transfer and hydrodynamic advancements are done separately and the information between the models is coupled explicitly.

$tt = 8$. NEARLY-IMPLICIT TIME ADVANCEMENT. If the fourth bit from the right is set (i.e., $tt = 8$ if the other bits are not set), the nearly-implicit scheme is used to advance the hydrodynamics; if the fourth bit from the right is not set, the semi-implicit scheme is used to advance the hydrodynamics.

$tt = 16$. STEADY-STATE CONVERGENCE NOT CHECKED. If the fifth bit from the right is set (i.e., $tt = 16$ if the other bits are not set), the test for convergence of a steady-state calculation is not made; if the fifth bit from the right is not set, the test for convergence of a steady-state calculation is made.

$tt = 32$. AUTOMATIC SELECTION BETWEEN SEMI- AND NEARLY-IMPLICIT TIME ADVANCEMENT. If the sixth bit from the right is set (i.e., $tt = 32$ if the other bits are not set), the on-line algorithm selection of time migration is used to advance the hydrodynamics. The semi-implicit scheme will be used when the time step is below the Courant limit and the nearly-implicit will be used when a large time step can be taken.

We recommend not using tt equal to 0 except for special testing situations. The use of tt equal to 1 is possible if the maximum time step is kept sufficiently small to ensure that the explicit connection between the heat conduction/transfer and hydrodynamics calculations remains stable. If there is any doubt, use tt equal to or greater than 3 (sets first bit and second bit). Using tt equal to 3 or 11 specifies the semi-implicit or the nearly-implicit advancement scheme, respectively, with both schemes using time step control, the heat conduction and hydrodynamics use the same time step, and the heat conduction/transfer and hydrodynamics are advanced separately. Using tt equal to 7 or 15 specifies the same features as tt equal to 3 or 11 and, in addition, specifies the implicit advancement of the heat conduction/transfer with the hydrodynamics. We recommend the nearly-implicit scheme during a steady-state and/or self-initialization case problem where the time step is limited by the material Courant limit. The nearly-implicit scheme can also be used during slower phases of a transient problem, though we advise the user that the answers may change somewhat from the semi-implicit scheme answers (depending on the time step size). The nearly-implicit advancement scheme is still under development and assessment; most of the verification and assessment for the code has been done with the semi-implicit advancement scheme. We did not recommend use of the implicit coupling of the heat conduction/transfer and hydrodynamics in prior versions since the implicit coupling was only partially implemented. With the implicit coupling now complete, we encourage the use of tt equal to 7 or 15. Users should be cautioned that the implicit coupling is a recent addition to RELAP5 and is still under assessment. When using the implicit coupling, the heat conduction time step must be the same as the hydrodynamic time step. This requirement is currently not enforced by the coding. In steady-state calculations, setting the fifth bit (adding 16) for the early part of the run can ensure the calculation runs to a user-specified time; then, setting the fifth bit off can allow the steady-state convergence to test control the termination of the problem.

- W5(I) Minor edit and plot frequency. This word has the packed format sssmmm. It is not necessary to input leading zeros. The mmm is the number of maximum or requested time advances per minor edit and the sss is the skip factor for the write of plot information. Thus $sss \times mmm$ is the number of maximum or requested time advances per plot-write on RSTPLT file. This feature enables to save the storage requirement for RSTPLT file. if no skip factor is set ($sss=0$), the default value is assigned, i.e. $sss=1$.
- W6(I) Major edit frequency. This is the number of requested time-advances per major edit.
- W7(I) Restart frequency. This is the number of requested time-advances per write of restart information.

4 MINOR EDITS, 301-399

CARDS 300, PLOT TIME SPAN

This card is optional for NEW and RESTART problems, are required for PC Window Plotting Case. If not entered, the initial time in CARD 2XX is set as a minimum value and final time is set as a maximum value.

W1(R) minimum X-value for X-Y Plot.

W2(R) Maximum X-Value for X-Y Plot.

CARDS 301 THROUGH 399, MINOR EDIT DATA REQUESTS

These cards are optional for NEW and RESTART problems, are required for a REEDIT problem, and are not allowed for PLOT and STRIP problems. If these cards are not present, no minor edits are printed. If these cards are present, minor edits are generated, and the order of the printed quantities is given by the card number of the request card. One request is entered per card, and the card numbers need not be consecutive. For RESTART problems, if these cards are entered, all the cards from the previous problem are deleted.

W1(A) Variable code (alphanumeric).

W2(I) Parameter (numeric).

W3(R) Minimum Y Value for X-Y Plot (Optional Select for Window Plotting).

W4(R) Maximum Y Value for X-Y Plot (Optional Select for Window Plotting).

W5(I) Window Identification for multiple X-Y Plot.

W6(I) Color Pallet Identification for multiple X-Y Plot.

Words 1 and 2 form the variable request code pair. The quantities that can be edited and the input required are listed below. For convenience, quantities that can be used in plotting requests, in trip specifications, as search variables in tables, and as operands in control statements are listed. Units for the quantities are also given. Interactive input variables described in Section 6 can be used in batch or interactive jobs in the same manner as the variables listed below. The parameter for interactive input variables is 1000000000. Quantities compared in variable trips must have the same units, and input to tables specified by variable request codes must have the specified units. The quantities are listed in alphabetical order within each section.

The underlined quantities without an asterisk in Section 4.1 through Section 4.8 are always written to the restart-plot file (RSTPLT). Underlined quantities followed by an asterisk have only some of the quantities written to the restart-plot file and the text will indicate which quantities are written.

4.1 General Quantities

<u>Variable</u>	<u>Description</u>
<u>COUNT</u>	Current attempted advancement count number. The parameter is 0.
<u>CPUTIME</u>	Current CPU time for this problem (s). The parameter is 0.
<u>DT</u>	Current time step (s). The parameter is 0.
<u>DTCRNT</u>	Current Courant time step (s). The parameter is 0.
<u>EMASS</u>	Estimate of mass error in all the systems (kg, lb). The parameter is 0.
ERRMAX	Current estimate of the truncation mass error fraction. The parameter is 0. This is the maximum of the two types of computed mass error (ϵ_m or ϵ_{rms}).
<u>EXTSNN</u>	Extra system variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “extra” flag on the dinstls command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for volumes called EXTVNN and for junctions called EXTJNN. Parameter is system number N.
NULL	Specifies null field. Allowed only on trip cards. The parameter is 0.
<u>RKTPOW3D</u>	Total reactor power for the MARS/MASTER kinetics coupled code (W). It is the sum of the fission and decay powers (W) for all the core heat structures. This variable is written to the restart-plot file only when card 1 option 88 is used.
STDTRN	Steady-state/transient flag. The parameter is 0. For steady-state, the value is 0.0. For transient, the value is 1.0.
YSERG	Total energy in system N (kJ, Btu). Parameter is system number N.
YSMER	Estimate of mass error in system n (kg, lb). Parameter is system number N.
YSTMS	Total mass of steam, water, and noncondensable in system N (kg, lb). Parameter is system number N.

TESTDA	An array testda, of twenty quantities, [REAL TESTDA(20)] has been defined for the convenience of program developers. This entry with a parameter ranging from 1 through 20 selects TESTDA(parameter). The testda array is initially set to 0.0, and programming must be inserted to set TESTDA values. The usual purpose of this capability is to allow a simple method for debug information to be printed in minor edits or to be plotted.
TIME	Time (s). The parameter is 0. This request cannot be used for minor edit requests.
TIMEOF	Time of trip occurring (s). The parameter is the trip number. This request is allowed only on trip cards.
<u>TMASS</u>	Total mass of water, steam, and noncondensables in all the systems (kg, lb). The parameter is 0.

4.2 Component Quantities

The quantities listed below are unique to certain components; for example, a pump velocity can only be requested for a pump component. The parameter is the component number, i.e., the three-digit number CCC used in the input cards.

<u>Variable</u>	<u>Description</u>
ACPGTG	Accumulator vapor specific heat, C_p , at vapor temperature (J/kg·K, Btu/lb·°F).
ACPNIT	Accumulator noncondensable specific heat, C_p , at vapor temperature (J/kg·K, Btu/lb·°F).
<u>ACQTANK</u>	Total energy transport to the gas by heat and mass transfer in the accumulator (W, Btu/s).
<u>ACRHON</u>	Accumulator noncondensable density (kg/m ³ , lb/ft ³).
<u>ACTTANK</u>	Mean accumulator tank wall metal temperature (K, °F).
<u>ACVDM</u>	Gas volume in the accumulator tank, standpipe, and surge line (m ³ , ft ³).
ACVGTG	Accumulator vapor specific heat, C_v , at vapor temperature (J/kg·K, Btu/lb·°F).
<u>ACVLIQ</u>	Liquid volume in the accumulator tank, standpipe, and surge line (m ³ , ft ³).
AHFGTF	Accumulator heat of vaporization at liquid temperature (J/kg, Btu/lb).
AHFGTG	Accumulator heat of vaporization at vapor temperature (J/kg, Btu/lb).

AHFTG	Accumulator liquid enthalpy at vapor temperature (J/kg, Btu/lb).
AHGTF	Accumulator vapor enthalpy at liquid temperature (J/kg, Btu/lb).
AVGTG	Accumulator specific volume at vapor temperature (m^3/kg , ft^3/lb).
AVISCN	Accumulator noncondensable viscosity ($\text{kg}/\text{m}\cdot\text{s}$, $\text{lb}/\text{ft}\cdot\text{s}$).
BETAV	Accumulator steam saturation coefficient of expansion (K^{-1} , $^{\circ}\text{F}^{-1}$).
CDIM	GE mechanistic dryer critical inlet moisture quality.
DIM	GE mechanistic dryer inlet moisture quality.
DMGDT	Accumulator/time rate of change in dome vapor mass (kg/s , lb/s).
GDRY	GE mechanistic separator capacity factor.
OMEGA	Inertial valve disk angular velocity (rad/s , rev/min).
<u>PMPHEAD</u>	Pump head in the pump component (Pa , lb_f/in^2).
PMPMT	Pump motor torque ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
PMPNRT	Calculated pump inertia ($\text{kg}\cdot\text{m}^2$, $\text{lb}\cdot\text{ft}^2$).
<u>PMPTRQ</u>	Pump torque in the pump component ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
<u>PMPVEL</u>	Pump rotational velocity in the pump component (rad/s , rev/min).
PRZLVL	Pressurizer level in the PRIZER component (m, ft).
THETA	Inertial valve disk angular position (deg).
<u>TUREFF</u>	Efficiency of the turbine component.
<u>TURPOW</u>	Power developed in the turbine component (W, Btu/s).
<u>TURTRQ</u>	Torque developed in the turbine component ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
<u>TURVEL</u>	Rotational velocity of the turbine component (rad/s , rev/min).

<u>VLVAREA</u>	Ratio of the current valve physical area to the junction area. The junction area is the fully open valve physical area for the smooth area option and the minimum of the two connecting volumes for the abrupt area change.
<u>VLVSTEM</u>	Ratio of the current valve stem position to the fully open valve stem position for the motor and servo valves when the normalized stem position option is used. For the motor and servo valves when the normalized area option is used and for all the other valves, this is the ratio of the current valve physical area to the fully open valve physical area.
XCO	GE mechanistic separator liquid carryover quality.
XCU	GE mechanistic separator vapor carryunder quality.
XI	GE mechanistic separator inlet quality.

4.3 Volume Quantities

For most of the following variable codes, the parameter is the volume number, i.e., the nine-digit number CCCNN0000 printed in the major edit. The parameter is CCC010000 for a single-volume; CCC010000 for a time-dependent volume; CCCNN0000 for a volume in a pipe component ($01 \leq NN \leq 99$); CCC010000 for the volume in a branch, separator, jetmixer, turbine, or ECC mixer component; CCC010000 for the volume in a pump component; and CCC010000 for the volume in an accumulator component. Some of the quantities are associated with the coordinate directions in the volume, and these quantities are computed for each coordinate direction in use. The parameter for the coordinate direction-related quantities is the volume number plus F, where F is described below. The quantities requiring the volume number plus F are so identified.

Every volume has at least one coordinate direction, and some volumes may have up to three orthogonal coordinate directions. Each coordinate has an inlet face and an outlet face. Faces are numbered 1 through 6, where faces 1 and 2 are the inlet and outlet faces associated with coordinate 1 (or x), respectively, faces 3 and 4 are inlet and outlet faces associated with coordinate 2 (or y), and faces 5 and 6 are inlet and outlet faces associated with coordinate 3 (or z). All volumes use coordinate 1. The quantity F to be added to the volume number to form the parameter used with coordinate direction related quantities is 0 or the face number. When F is 0 (i.e., just the volume number) or 1 or 2, the volume velocity is for coordinate 1. When F is 3 or 4, the volume velocity is for coordinate 2, and when F is 5 or 6, the volume velocity is for coordinate 3. For the underlined quantities followed by an asterisk in the list below, the coordinate-dependent quantities for coordinate 1 are automatically written to the restart-plot records using the parameter with F equal to 0. Input checks are made to ensure the parameter specifies a volume coordinate direction that is in use.

<u>Variable</u>	<u>Description</u>
AVOL	Area of the volume (m^2 , ft^2); the parameter is the volume number plus F.

BETAFF	Liquid isobaric coefficient of the thermal expansion, β_f , bulk conditions (K^{-1} , $^{\circ}F^{-1}$).
BETAGG	Vapor isobaric coefficient of the thermal expansion, β_g , bulk conditions (K^{-1} , $^{\circ}F^{-1}$).
<u>BORON</u>	Spatial boron density, ρ_b (kg/m^3 , lb/ft^3). This is the volume liquid fraction (α_f) times the liquid density (ρ_f) times the boron concentration (C_b). Boron concentration is used for hydrodynamic input, and boron density is used for minor edits and plots.
BRNPPM	Boron concentration in ppm unit.
CSUBPF	Liquid specific heat, C_{pf} , bulk conditions ($J/kg\cdot K$, $Btu/lb\cdot^{\circ}F$).
CSUBPG	Vapor specific heat, C_{pg} , bulk conditions ($J/kg\cdot K$, $Btu/lb\cdot^{\circ}F$).
DRFDP	Partial derivative of ρ_f with respect to P (s^2/m^2 , s^2/ft^2).
DRFDUF	Partial derivative of ρ_f with respect to U_f ($kg\cdot s^2/m^5$, $lb\cdot s^2/ft^5$).
DRGDP	Partial derivative of ρ_g with respect to P (s^2/m^2 , s^2/ft^2).
DRGDUG	Partial derivative of ρ_g with respect to U_g ($kg\cdot s^2/m^5$, $lb\cdot s^2/ft^5$).
DRGDXA	Partial derivative of ρ_g with respect to X_n (kg/m^3 , lb/ft^3).
DTDP	Partial derivative of T^s with respect to P (K/Pa , $in^2\cdot^{\circ}F/lb_f$).
DTDUG	Partial derivative of T^s with respect to U_g ($s^2\cdot K/m^2$, $s^2\cdot^{\circ}F/ft^2$).
DTDXA	Partial derivative of T^s with respect to X_n (K , $^{\circ}F$).
DTFDP	Partial derivative of T_f with respect to pressure (K/Pa , $in^2\cdot^{\circ}F/lb_f$).
DTFDUF	Partial derivative of T_f with respect to U_f ($s^2\cdot K/m^2$, $s^2\cdot^{\circ}F/ft^2$).
DTGDP	Partial derivative of T_g with respect to P (K/Pa , $in^2\cdot^{\circ}F/lb_f$).
DTGDUG	Partial derivative of T_g with respect to U_g ($s^2\cdot K/m^2$, $s^2\cdot^{\circ}F/ft^2$).

DTGDXA	Partial derivative of T_g with respect to X_n (K, °F).
<u>EXTVNN</u>	Extra volume variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “extra” flag on the dinstls command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for junctions called EXTJNN and for systems called EXTSNN.
<u>FLOREG</u>	Flow regime number; the parameter is the volume number. A chart showing the meaning of each number is shown in Volume I, Section 3.1.3.
FWALF	Liquid wall frictional drag coefficient ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$); the parameter is the volume number plus F.
FWALG	Vapor wall frictional drag coefficient ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$); the parameter is the volume number plus F.
GAMMAC	For explicit coupling of heat conduction/transfer and hydrodynamics, this is 0. For implicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
GAMMAI	Mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
GAMMAW	For explicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation/condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$). For implicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
HGF	Direct heating heat transfer coefficient per unit volume ($\text{W/m}^3\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^3\cdot^\circ\text{F}$).
HIF	Liquid side interfacial heat transfer coefficient per unit volume ($\text{W/m}^3\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^3\cdot^\circ\text{F}$).
HIG	Vapor side interfacial heat transfer coefficient per unit volume ($\text{W/m}^3\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^3\cdot^\circ\text{F}$).
HSTEAM	Steam specific enthalpy at bulk conditions using partial pressure of steam (J/kg, Btu/lb).
HVMIX	Enthalpy of the liquid and vapor (J/kg, Btu/lb).

<u>P</u>	Volume pressure (Pa, lb _f /in ²).
PECLTV	Peclet number.
PPS	Steam partial pressure (Pa, lb _f /in ²).
Q	Total volume heat source from the wall and direct moderator heating to liquid and vapor (W, Btu/s). This variable request is the same as “Q.wall.tot.” in the major edits.
<u>QUALA</u>	Volume noncondensable mass fraction. The ratio of the mass of the noncondensable gas to the total mass of the vapor phase.
<u>QUALE</u>	Volume equilibrium quality. This quality uses phasic enthalpies and mixture quality, with the mixture enthalpy calculated using the flow quality.
<u>QUALS</u>	Volume static quality.
<u>QWG</u>	Volume heat source from the wall and direct moderator heating to vapor (W, Btu/s). This variable request is the same as “Qwg.wall.gas.” in the major edits.
<u>RHO</u>	Total density (kg/m ³ , lb/ft ³).
<u>RHOF</u>	Liquid density ρ_f (kg/m ³ , lb/ft ³).
<u>RHOG</u>	Vapor density ρ_g (kg/m ³ , lb/ft ³).
RHOM	Total density for the mass error check (kg/m ³ , lb/ft ³).
SATHF	Liquid specific enthalpy at saturation conditions using partial pressure of steam (J/kg, Btu/lb).
SATHG	Steam specific enthalpy at saturation conditions using partial pressure of steam (J/kg, Btu/lb).
<u>SATTEMP</u>	Volume saturation temperature based on the partial pressure of steam (K, °F).
SIGMA	Surface tension (N/m, lb _f /ft).
<u>SOUNDE</u>	Volume sonic velocity (m/s, ft/s).
<u>TEMPF</u>	Volume liquid temperature T_f (K, °F).

<u>TEMPG</u>	Volume vapor temperature T_g (K, °F).
TFSUB	Degree of subcooling for volume liquid (K, °F), <0.0 .
TGSUP	Degree of superheating for volume vapor (K, °F), >0.0 .
THCONF	Liquid thermal conductivity (W/m•K, Btu/s•ft•°F).
THCONG	Vapor thermal conductivity (W/m•K, Btu/s•ft •°F).
TIENGV	Total internal energy (of both phases and noncondensables) in volume (J, Btu).
TMASSV	Total mass (includes both phases and noncondensables) in volume (kg, lb).
TSATT	Saturation temperature corresponding to total pressure (K, °F).
<u>UF</u>	Liquid specific internal energy (J/kg, Btu/lb).
<u>UG</u>	Vapor specific internal energy (J/kg, Btu/lb).
<u>VAPGEN</u>	Total mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation and in the boundary layer near the wall for vapor generation/condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
<u>VELF*</u>	Volume oriented liquid velocity (m/s, ft/s); the parameter is the volume number plus F.
<u>VELG*</u>	Volume oriented vapor velocity (m/s, ft/s); the parameter is the volume number plus F.
VISCF	Liquid viscosity ($\text{kg/m}\cdot\text{s}$, $\text{lb/ft}\cdot\text{s}$).
VISCG	Vapor viscosity ($\text{kg/m}\cdot\text{s}$, $\text{lb/ft}\cdot\text{s}$).
<u>VOIDF</u>	Volume liquid fraction.
<u>VOIDG</u>	Volume vapor fraction (void fraction).
VOIDLA	Void above the level.
VOIDLB	Void below the level.
VOLLEV	Location of the level inside the volume (m, ft).
VVOL	Volume of the volume (m^3 , ft^3).

4.4 Junction Quantities

For the following variable request codes, the parameter is the junction number, i.e., the nine-digit number CCCNN0000 printed in the major edit. The parameter is CCC000000 for a single-junction; CCC000000 for a time-dependent junction; CCCMM0000 for a junction in a pipe component ($01 \leq MM \leq 99$); CCCMM0000 for a junction in a branch, separator, jetmixer, turbine, or ECC mixer component ($01 < MM \leq 9$); CCC000000 for a valve junction; CCC010000 for the inlet junction in a pump component; CCC020000 for the outlet junction in a pump component; CCCIINN00 for a junction in the multiple-junction component ($01 \leq II \leq 99$, $01 \leq NN \leq 99$); and CCC010000 for the junction in an accumulator component.

<u>Variable</u>	<u>Description</u>
C0J	Junction distribution coefficient. The 0 in C0J is the number 0 and not the upper case letter O.
CHOKEF	Junction choking flag. The value is 0 if the flow is not choked, and is 1 if the flow is choked.
<u>EXTJNN</u>	Extra junction variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “extra” flag on the dinstls command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for volumes called EXTVNN and for systems called EXTSNN.
FIJ	Interphase friction coefficient ($\text{N}\cdot\text{s}^2/\text{m}^5$, $\text{lb}_\text{f}\cdot\text{s}^2/\text{ft}^5$).
FJUNFT	Total forward user input form loss coefficient for irreversible losses, including Re dependence (dimensionless).
FJUNRT	Total reverse user input form loss coefficient for irreversible losses, including Re dependence (dimensionless).
FLENTH	Total enthalpy flow in junction (includes both phases and noncondensables) (J/s, Btu/s).
FLORGJ	Junction flow regime number. A chart showing the meaning of each number is shown in Volume I, Section 3.
FORMFJ	Liquid abrupt area change model form loss factor (dimensionless).
FORMGJ	Vapor abrupt area change model form loss factor (dimensionless).
FWALFJ	Non-dimensional liquid wall friction coefficient (dimensionless).

FWALGJ	Non-dimensional vapor wall friction coefficient (dimensionless).
IREGJ	Vertical bubbly/slug flow junction flow regime number. A chart showing the meaning of each number is shown in Volume I, Section 3.
<u>MFLOWJ</u>	Combined liquid and vapor flow rate (kg/s, lb/s).
MFLOWFJ	Liquid flow rate (kg/s, lb/s).
MFLOWGJ	Vapor flow rate (kg/s, lb/s).
MFLOWNJ	Noncondensable gas flow rate (kg/s, lb/s).
<u>QUALAJ</u>	Junction noncondensable mass fraction.
<u>RHOEJ</u>	Junction liquid density (kg/m^3 , lb/ft^3).
<u>RHOGJ</u>	Junction vapor density (kg/m^3 , lb/ft^3).
SONICJ	Junction sound speed (m/s, ft/s). This speed is based on the physical junction area. It does not include the effects of the throat ratio and the discharge coefficients.
<u>UFJ</u>	Junction liquid specific internal energy (J/kg, Btu/lb).
<u>UGJ</u>	Junction vapor specific internal energy (J/kg, Btu/lb).
<u>VELFJ</u>	Junction liquid velocity (m/s, ft/s). This velocity is based on the junction area A_j .
<u>VELGJ</u>	Junction vapor velocity (m/s, ft/s). This velocity is based on the junction area A_j .
VGJJ	Vapor drift velocity (m/s, ft/s).
<u>VOIDFJ</u>	Junction liquid fraction.
<u>VOIDGJ</u>	Junction vapor fraction (void fraction).
VOIDJ	Junction vapor fraction (void fraction) used in the interphase friction.
XEJ	Junction equilibrium quality.

4.5 Heat Structure Quantities

For the request code, HTVAT, the parameter is the seven-digit heat structure number CCCG0NN. For the remaining codes, the parameter is the seven-digit heat structure number CCCG0NN with a two-digit number appended. For codes other than HTTEMP and HTVAT, the appended number is 00 for the left boundary and 01 for the right boundary. For HTTEMP, the appended number is the mesh point number. For HTVAT, omit the two appended digits and use only the seven digit number. Only the left and right surface temperatures are written by default in plot records on the RSTPLT file, and thus, plot requests in plot-type problems and strip requests are limited to those temperatures

<u>Variable</u>	<u>Description</u>
GAPPRES	Gap pressure (Pa).
GAPCOND	Gap effective conductivity (W/m-K, Btu/s-ft-R)
<u>HTCHF</u>	Critical (maximum) heat flux (W/m^2 , Btu/s-ft ²).
<u>HTCHFR</u>	Critical hat flux ratio (ratio of HTCHF to HTRNR). This variable can only be plotted and is set equal to the constant value of 12.34567 whenever HTCHF or HTRNR is equal to 0.0.
HTGAMW	Wall vapor generation rate per unit volume ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$). The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
<u>HTHTC</u>	Heat transfer coefficient ($\text{W/m}^2\cdot\text{K}$, Btu/s-ft ² ·°F).
HTMODE	Boundary heat transfer mode number (unitless). The mode number indicates which heat transfer regime is currently in effect. The parameter is the seven-digit heat structure geometry number, CCCG0NN, with a two-digit number appended. The two-digit appended number, 00, specifies the left boundary, and 01 specifies the right boundary. This same quantity is valid for the reflood heat structures.
HTRG	Heat flux to vapor phase (W/m^2 , Btu/s-ft ²). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
<u>HTRNR</u>	Heat flux (W/m^2 , Btu/s-ft ²).
<u>HTTEMP</u>	Mesh point temperature (K, °F). The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (mesh point number). The surface

temperatures are written to the plot record but interior mesh point temperatures must be requested through the 2080XXXX cards.

<u>HTVAT</u>	Volume averaged temperature in the heat structure (K, °F).
PECL	Liquid Peclet number for the heat structures. The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
QRAD	Heat flux (W/m^2 , Btu/s.ft^2) due to radiation and contact heat transfer. The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
STANT	Stanton number. The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).

4.6 Reflood-Related Quantities

For the following variable codes, the parameter is the heat structure geometry number, i.e., the four-digit number CCCG printed in the major edit.

<u>Variable</u>	<u>Description</u>
FINES	Current number of axial nodes on a reflood structure.
TCHFQF	Temperature at the critical (maximum) heat flux (K, °F).
TREWET	Rewet, quench, Leidenfrost, or minimum film boiling temperature (K, °F).
ZQBOT	Elevation of bottom quench front (m, ft).
ZQTOP	Elevation of top quench front (m, ft).

4.7 Reactor Kinetic Quantities

The parameter is 0 for the following reactor kinetic quantities.

<u>Variable</u>	<u>Description</u>
REAC	Reactivity feedback total (dollars). This is the sum of REACM, REACRB, REACS, and REACTF.
RKBORN	Reactivity feedback from boron density changes (dollars).

RKDOPP	Reactivity feedback from fuel temperature changes (dollars).
RKMOD	Reactivity feedback total from moderator (dollars). This is the sum of RKMDD and RKMOT.
RKMDD	Reactivity feedback from moderator density changes (dollars).
RKMOT	Reactivity feedback from moderator temperature (spectral) changes (dollars).
RKSCRAM	Reactivity feedback from scram curve (dollars).
RKFIPOW	Reactor power from fission (W).
<u>RKGAPOW</u>	Reactor power from decay of fission products and actinides (W).
RKPOWA	Reactor power from decay of actinides (W).
<u>RKPOWK</u>	Reactor power from decay of fission products (W).
<u>RKREAC</u>	Reactivity (dollars).
<u>RKRECPER</u>	Reciprocal period (s^{-1}).
<u>RKTPOW</u>	Total reactor power, i.e., sum of fission and decay powers (W).

4.8 Control System Quantities

The parameter is the control component number, i.e., the three-digit number, CCC, or the four-digit number, CCCC, used in the input cards.

<u>CNTRLVAR</u>	Control component number. These quantities are assumed dimensionless except for a SHAFT component.
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5 TRIPS, 400-799 or 20600000-20620000

These cards are optional for NEW and RESTART type problems and are not used for other problem types. Two different card series are available for entering trip data, but only one series type may be used in a problem. Card numbers 401-799 allow 199 variable trips and 199 logical trips. Card numbers 20600010 - 20620000 allow 1000 variable trips and 1000 logical trips.

5.1 Trips Cancellation, 400

This card is allowed only for RESTART problems. The card causes all trips in the problem being restarted to be deleted. Any desired trips must be reentered.

W1(A) DISCARD. Any other entry is an error.

5.2 Trip Series Type, 20600000

This card, if omitted, selects card numbers 401-599 for variable trips and 601-799 for logical trips. For this case, the trip numbers are equal to the card numbers.

If this card is entered, card numbers 206NNNN0 are used for entering trip data, where NNNN is the trip number. Trip numbers (NNNN) 1-1,000 are variable trips, and 1,001-2,000 are logical trips. Trip numbers do not have to be consecutive.

W1(A) EXPANDED. Any other entry is an error.

5.3 Variable Trips, 401-599 or 20600010-20610000

Each card defines a logical statement or trip condition concerned with the quantities being advanced in time. A trip is false or not set if the trip condition is not met, and true if it is met. On restart, new trips can be introduced, old trips can be deleted, and a new trip with the same number as an old trip replaces the old trip.

The variables and parameters are the same as described for minor edits, Section 4. NULL is allowed for the right side when only a comparison to the constant is desired. The variable code TIMEOF, with the parameter set to the trip number, indicates the time at which the trip was last set. If the trip goes false, TIMEOF is set to -1.0.

W1(A) Variable. On RESTART problems, this word can also contain DISCARD or RESET. DISCARD deletes the trip; RESET sets the trip to false. If DISCARD or RESET are entered, no further words are entered on the card.

W2(I) Parameter.

W3(A)	Relationship. This may be either EQ, NE, GT, GE, LT, or LE, where the symbols have the standard FORTRAN meaning. Do not enter periods as part of the designator. For example, use GE rather than “.GE.” to specify <i>greater than</i> or <i>equal to</i> .
W4(A)	Variable.
W5(I)	Parameter.
W6(R)	Additive constant.
W7(A)	Latch indicator. If L, the trip once set true remains true, even if the condition later is not met. If N, the trip is tested each time advancement.
W8(R)	Timeof quantity (s). This word is optional. If it is not entered, the trip is initialized as false and the associated TIMEOF quantity is set to -1.0. If -1.0 is entered, the trip is initialized as false. If 0 or a positive number is entered for TIMEOF, the trip is initialized as true. TIMEOF must not be greater than 0 for NEW problems and must not be greater than the time of restart for RESTART problems.
W9(A)	Auxiliary Input for Trip Message. This word is optional. If it is not entered, no message will be displayed. It must not greater than 24 character.

The logical statement is “Does the quantity given by Words 1 and 2 have the relationship given by Word 3 with the quantity given by Words 4 and 5 plus Word 6?” If the relationship is false, the trip is false or not set. If the relationship is true, the trip is true or set. The TIMEOF variable is -1.0 if the trip is false. If the trip is true, this variable is the time the trip was last set true. A latched trip is never reset, so the trip time never changes once it changes from -1.0. For the nonlatched trips, the trip time when set remains constant until the trip condition becomes false and then the trip time is -1.0 again. If the trip condition becomes true again, the process is repeated. For trips such as a time test, L should be used to eliminate repeated testing, although no error or difference in results will occur if N is used.

5.4 Logical Trips, 601-799 or 20610010-20620000

If these cards are entered, at least one of the variable trips must have been entered. Each card defines a logical relationship with the trips defined on these cards or on the variable trips.

W1(I)	Trip number. The absolute value of this number must be one of the trip numbers defined by the variable or logical trips. A negative trip number indicates that the complement of the trip is to be used in the test.
W2(A)	Operator. The operator may be AND, OR, or XOR. For RESTART problems, this quantity may also contain DISCARD or RESET. DISCARD deletes the trip and RESET

sets the trip to false. If DISCARD or RESET are entered, no further words are entered on the card and Word 1 (W1) may be 0.

- W3(I) Trip number. This is similar to Word 1 (W1).
- W4(A) Latch indicator. If L, the trip when set remains set. If N, the trip is tested each time advancement.
- W5(R) Timeof quantity (s). This word is optional. If not entered, the trip is initialized as false, and the associated TIMEOF quantity is set to -1.0. If -1.0 is entered, the trip is initialized as false. If 0 or a positive number is entered for TIMEOF, the trip is initialized as true. TIMEOF must not be greater than 0 for NEW problems and must not be greater than the time of restart for RESTART problems.
- W6(A) Auxiliary Input for Trip Message. This word is optional. If it is not entered, no message will be displayed. It must not greater than 24 character.

The trip condition is given by the result of the following logical expression:

CONDITION OF TRIP IN W1 OPERATOR CONDITION OF TRIP IN W3.

5.5 Trip Stop Advancement, 600

This card can be entered in NEW and RESTART problems. One or two trip numbers may be entered. If either of the indicated trips are true, the problem advancement is terminated. These trips are tested only at the end of a requested advancement. If the trips can cycle true and false, they should be latched-type trips to ensure being true at the test time.

- W1(I) Trip number.
- W2(I) Trip number. A second trip number need not be entered.

6 INTERACTIVE INPUT, 801-999

6.1 CARDS 801 THROUGH 999, INTERACTIVE INPUT DATA

An interactive control function is available with the Nuclear Plant Analyzer (NPA) version. This function allows a user to interactively control the trip logic, valve area, mass flow rate of the time-dependent junctions, and heater power.

These data may be entered for batch and interactive jobs. However, the interactive control input data have no effects in the case of batch job. In the case of interactive jobs, an interactive control window is provided for the interactive control. The interactive control input data cards can be used in a NEW or RESTART job. In a restart job, these data can be added (cannot be replaced or removed).

These cards define variables that can be interactively changed during execution by user. The card input defines input variable names (variable code and parameter) and comments.

- | | |
|-------|--|
| W1(A) | Type of interactive control (alphanumeric). Enter one of the following interactive control types; "trip" for trip control, "vlvarea" for servo valve area control, "mflowfj" for liquid flow rate control of a time-dependent junction, "mflowgj" for vapor flow rate control of a time-dependent junction, and "power" for heater power control. |
| W2(I) | Parameter (numeric). The parameter is defined according to the type of interactive control as follows: trip card number (400 through 799) for trip control, valve component number (ccc) for servo valve area control, time-dependent junction number (ccc) for liquid flow rate control, time-dependent junction number (ccc) for vapor flow rate control, and general table number (ttt) for heater power control. |
| W3(A) | Comments (alphanumeric). Enter comments for the interactive control. The comment should not be more than 32 characters including quotation marks. These comments are displayed in the interactive control window. |

It is noted that relevant input data should exist for interactive control. That is, for interactive trip control, the trip input should be entered. The value of the trip is determined by user-specified trip input if user action is not given, but upon the user action the value is manually changed. In the same manner, a servo valve component input should be entered for valve area control, and a time-dependent junction component input should be entered for mass flow rate control. For heater power control, the general table input for "power" should be prepared.

7 STRIP REQUESTS, 1001-1999

These cards are required only in STRIP-type problems. One or more cards are entered, each card containing one variable request. Card numbers need not be consecutive. Variables are ordered on the STRIPF file in the order of increasing card numbers. If an incorrect variable request code is entered, the value will be 0.0. It is not flagged as an input error, since at some later time in the transient, a renodalization may result in the variable request code becoming correct.

7.1 CARDS 1000, X-VALUE DATA FOR PLOTTING

W1(R) Minimum X-Value for Window X-Y Plot

W2(R) Maximum X-Value for Window X-Y Plot

7.2 CARDS 1001 THROUGH 1999, STRIP REQUEST DATA

These cards are required only in STRIP-type problems. One or more cards are entered, each card containing one variable request. Card numbers need not be consecutive. Variables are ordered on the STRIPF file in the order of increasing card numbers. If an incorrect variable request code is entered, the value will be 0.0. It is not flagged as an input error, since at some later time in the transient, a renodalization may result in the variable request code becoming correct.

W1(A) Alphanumeric part of the variable request code.

W2(I) Integer part of the variable request code.

W3(R) Minimum Y-Value for Window X-Y Plot

W4(R) Maximum Y-Value for Window X-Y Plot

